

Roll Number

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**ODD END SEMESTER EXAMINATION DECEMBER-2024**

Name of the Course: B.Tech

Semester: 3<sup>rd</sup>

Name of the Paper: Electrical and Electronics measuring instruments

Paper Code: TEE 303

Time: 3 Hours

Maximum Marks: 100

Note:-

- I. All Questions are compulsory.
- II. Answer any two sub questions among a, b & c in each main question.
- III. Total marks in each question are twenty.
- IV. Each question carries 10 marks.

Q1. Attempt any Two

(10\*2=20)

- a) What are the terms measurements and measurement system? Describe the functional element of measurement system with block diagram. (CO1)
- b) Differentiate between electrical, electronic and mechanical instruments with suitable examples. (CO1)
- c) Explain the following terms: (CO1)
  - I. Accuracy
  - II. Selectivity
  - III. Sensitivity
  - IV. Errors

Q2. Attempt any TWO

(10\*2=20)

- a) Discuss briefly about bridges? Draw the Schering Bridge and derive necessary equations & phasor. (CO2)
- b) A Maxwell bridge is used to measure an inductive impedance the bridge constants at balance are  $C1=0.05 \mu F$ ,  $R1=500 K\Omega$ ,  $R2=10.1 K\Omega$  and  $R3=10 K\Omega$ . Find the series equivalent of the unknown impedance (CO2)
- c) How can we increase the range of Ammeter and Voltmeter? (CO2)

Q3. Attempt any TWO

(10\*2=20)

- a) Draw the block diagram of a Cathode ray oscilloscope and explain function of each block in detail. (CO5)
- b) Explain the major parts of Cathode ray tube with a block diagram. Compare, explain different types of CRO. (CO5)
- c) How we can measure the frequency and phase from CRO (CO5)

Q4. Attempt any TWO

(10\*2=20)

- a) Explain Ramp and Dual slope voltmeter with block diagram. (CO4)
- b) What is difference between analog voltmeter and digital voltmeter. (CO4)
- c) Write a short note on flip-flop, Registers and counters. (CO4)

Q5. Attempt any TWO

(10\*2=20)

- a) Explain Different types of instrument transformer and its error (CO3)
- b) Explain PMMC and MI instruments with a neat and clean diagram. (CO2)
- c) Why Kelvin's bridge is used explain with the diagram. (CO2)